

IEAP-Rstudio-Series03: Data wrangling and ANOVA

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1 ANOVA: Analysis of variance

We want to reproduce Fig 3 of the article: the effect of difficulty (ID) on movement time (MT) in three groups (ap, cp, pp).

Section: 1.1

1. **Is the effect of difficulty the same in all groups?** We look at the **GROUP x ID interaction**. If it is significant, the effect of difficulty is different between groups (for example, much bigger in one group than in the others). If it is not significant, difficulty has about the same effect in all groups.
2. **How do we visualize this?** We plot movement time (MT, y axis) as a function of difficulty (ID, x axis), with one line per group. If the lines are parallel, the effect is the same in all groups. If the slopes are different, the effect is different. The slope of each regression line tells us how many seconds MT increases when ID increases by one bit.
3. **Is there a significant difference?** We read the p-values of the ANOVA. If $p < .05$, the difference is unlikely to be due to chance. We look at three effects: GROUP (do the groups differ overall?), ID (does difficulty change MT?) and GROUP x ID (does difficulty change MT differently in each group?).
4. **What sentence do we write in the report?** We write one short sentence for each result of the ANOVA, with its numbers in brackets. The numbers are F, the degrees of freedom and the p-value. For example: "Movement time increased with difficulty (F = ..., p < ...). This increase was different between the groups (F = ..., p < ...)." The final sentence with our real numbers is in section 1.3.

GROUP is a between-subject factor and ID is a within-subject factor, so we need a **mixed ANOVA**.

1.2 What library to answer the above questions?

- tidyverse to reorganize the data and plot
- ez to run the ANOVA

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.3.3

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.1.5
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ez)
```

Warning in check_dep_version(): ABI version mismatch:
lme4 was built with Matrix ABI version 1
Current Matrix ABI version is 0
Please re-install lme4 from source or restore original 'Matrix' package

1.3 Data reorganization

```
df <- read_table("data/Results.txt", show_col_types = FALSE) %>%
  rename(MT = MovementTime) %>%
  mutate(GROUP = factor(GROUP), SUBJ = factor(SUBJ),
         ID = factor(ID), REP = factor(REP)) %>%
  as.data.frame()
```

Count the data points for each combination of factors, to look for missing data.

```
df %>% count(GROUP, ID)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	ap	ID3	740
4	cp	ID1	1140
5	cp	ID2	1140
6	cp	ID3	1140
7	pp	ID1	760
8	pp	ID2	760
9	pp	ID3	400

Data for groups is imbalanced for ID3 in the group pp. We remove ID3.

```
df <- df %>% filter(ID != "ID3") %>% droplevels()
levels(df$ID)
```

```
[1] "ID1" "ID2"
```

```
df %>% count(GROUP, ID)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	cp	ID1	1140
4	cp	ID2	1140
5	pp	ID1	760
6	pp	ID2	760

The data are still unbalanced. Let's check REP.

```
df %>% count(REP)
```

	REP	n
1	R1	2040
2	R2	2040
3	R3	1200

```
df %>% count(GROUP, REP)
```

	GROUP	REP	n
1	ap	R1	520
2	ap	R2	520
3	ap	R3	440
4	cp	R1	760
5	cp	R2	760
6	cp	R3	760
7	pp	R1	760
8	pp	R2	760

Data imbalanced for REP3 in the groups pp and ap. We remove REP3.

```
df <- df %>% filter(REP != "R3") %>% droplevels()
df %>% count(GROUP, REP)
```

	GROUP	REP	n
1	ap	R1	520
2	ap	R2	520
3	cp	R1	760
4	cp	R2	760
5	pp	R1	760
6	pp	R2	760

Number of participants per group:

```
df %>% distinct(GROUP, SUBJ) %>% count(GROUP, name = "count") %>% as.data.frame()
```

	GROUP	count
1	ap	13
2	cp	19
3	pp	19

One value per participant and ID

The ANOVA wants a single value per participant and per condition: we make the average.

```
data_anova <- df %>%
  group_by(GROUP, SUBJ, ID) %>%
  summarise(MT = mean(MT), .groups = "drop")
```

ANOVA

```
anova_res <- ezANOVA(data = data_anova,
  dv = MT,
  wid = SUBJ,
  within = ID,
  between = GROUP,
  detailed = FALSE)
```

Warning: Data is unbalanced (unequal N per group). Make sure you specified a well-considered value for the type argument to ezANOVA().

```
anova_res$ANOVA
```

	Effect	DFn	DFd	F	p	p<.05	ges
2	GROUP	2	48	37.24408	1.720007e-10	*	0.57462110
3	ID	1	48	30.77618	1.221240e-06	*	0.07667596
4	GROUP:ID	2	48	11.58124	7.866206e-05	*	0.05882293

Movement time increased with difficulty, $F(1, 48) = 30.78$, $p < .001$. This increase was different between the groups, $F(2, 48) = 11.58$, $p < .001$.

1.4 Run the linear regression by GROUP

```
data_reg <- data_anova %>%
  mutate(ID_num = parse_number(as.character(ID)))

reg_table <- data_reg %>%
  group_by(GROUP) %>%
  group_modify(~ {
    m <- lm(MT ~ ID_num, data = .x)
    tibble(intercept = coef(m)[1], slope = coef(m)[2])
  }) %>%
```

```
mutate(label = sprintf("MT = %.3f + %.3f * ID", intercept, slope))

reg_table
```

```
# A tibble: 3 x 4
# Groups:   GROUP [3]
  GROUP intercept slope label
<fct>      <dbl> <dbl> <chr>
1 ap         0.730 0.269 MT = 0.730 + 0.269 * ID
2 cp         0.634 0.201 MT = 0.634 + 0.201 * ID
3 pp         1.69  1.46  MT = 1.689 + 1.464 * ID
```

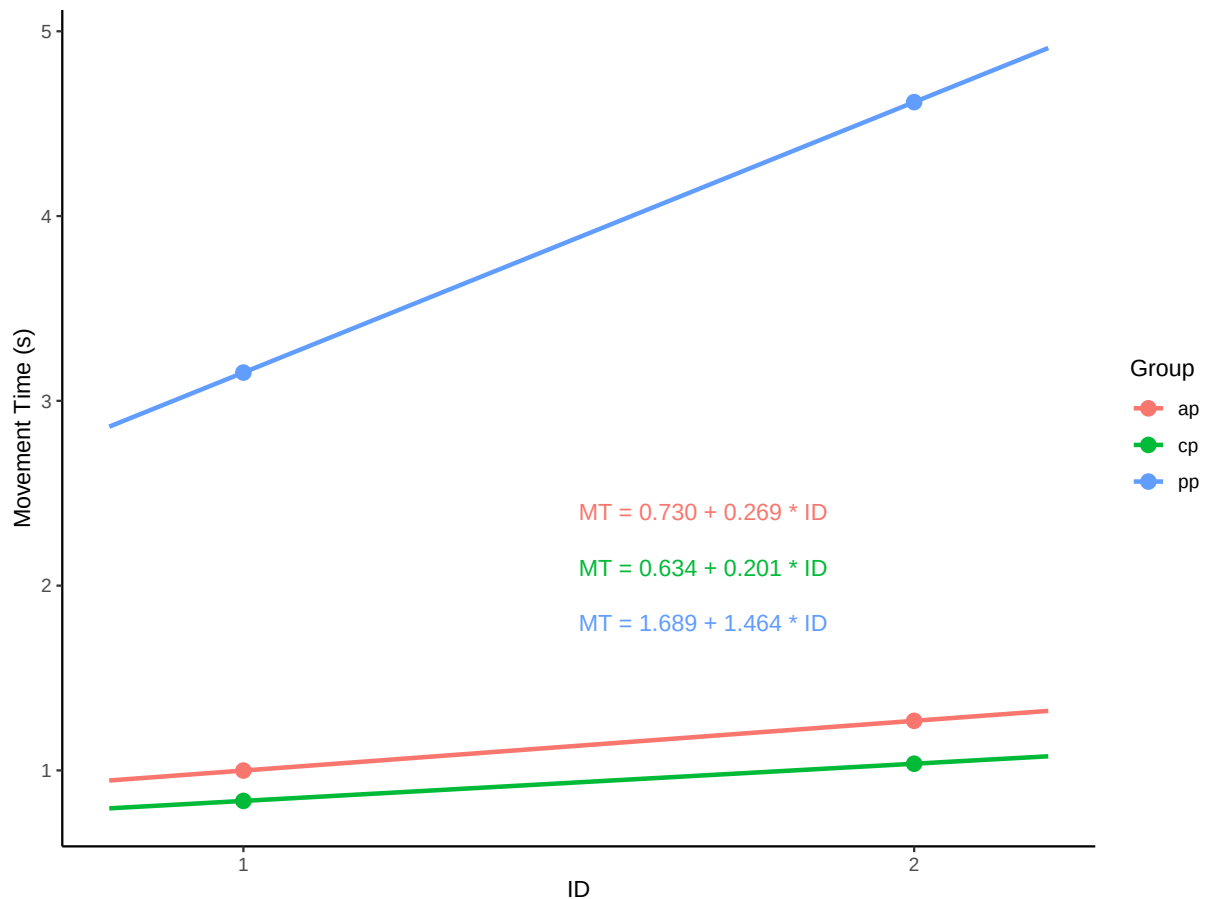
1.5 Graph for publication

```
means <- data_reg %>%
  group_by(GROUP, ID_num) %>%
  summarise(MT = mean(MT), .groups = "drop")

eq_pos <- reg_table %>%
  ungroup() %>%
  mutate(x = 1.5, y = 2.7 - 0.3 * row_number())

p <- ggplot(means, aes(x = ID_num, y = MT, colour = GROUP)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
             fullrange = TRUE) +
  geom_text(data = eq_pos, aes(x = x, y = y, label = label),
           hjust = 0, show.legend = FALSE) +
  scale_x_continuous(breaks = c(1, 2), limits = c(0.8, 2.2)) +
  labs(x = "ID", y = "Movement Time (s)", colour = "Group") +
  theme_classic()
```

p



1.6 Export the graph as a PDF

```
dir.create("figures", showWarnings = FALSE)
ggsave("figures/fig3_MT_ID.pdf", plot = p, width = 8, height = 6,
       units = "in")
```

1.7 Discuss the differences with the article

Our regression equations (e.g., $MT = 0.730 + 0.269 \times ID$ for ap) differ from those in the article. Our method is simpler: we averaged the data per participant and per ID, then ran a classic ANOVA. Nevertheless, the $GROUP \times ID$ interaction is significant, $F(2, 48) = 11.58$, $p < .001$, so the conclusion is the same: the effect of difficulty depends on the group. The simple method keeps the main pattern of the data: a much steeper increase of movement time with ID in the

pp group than in the ap and cp groups. The more sophisticated method of the article changes the exact values (slopes and intercepts), but not this pattern.